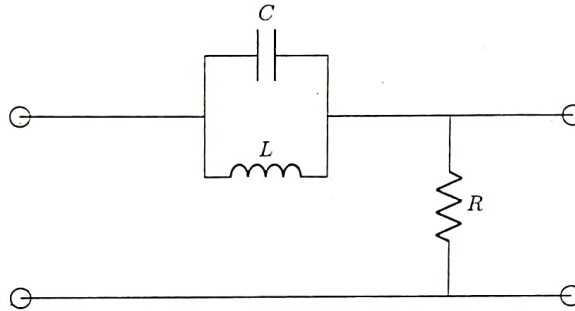


## PH3104 : Electrical Circuits and Electronics

### Mid Semester Exam

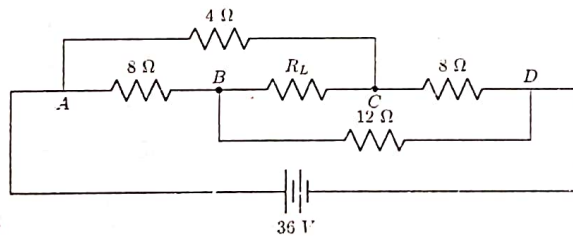
Answer all of the following questions. Attach the open sheet to the back of your answer scripts

1. We derived the The Lumped Circuit/Element model in class from Maxwell's equations. The Lumped Circuit Model is typically better suited for signals in which frequency ranges :  
 A. DC   B. Terahertz   C. Kilohertz   D. Gigahertz  
 Give the correct option(s) with proper Justification. [2]
2. Approximately plot the voltage across the load resistance as a function of frequency for the following circuit. At what frequency will the maximum or minimum voltage occur.



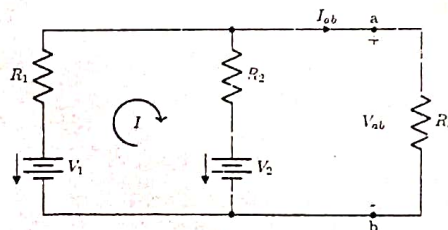
[3]

3. Using Thevenin's Theorem in the following figure, find  $V_{Th}$ ,  $R_{Th}$  across the points B and C and the current flowing through load  $R_L = 6\Omega$ .



[4]

4. Draw the Norton equivalent for the network to the left of the terminals in the following circuit, assuming  $V_1 = 10V$ ,  $V_2 = 15V$ ,  $R_1 = 4\Omega$  and  $R_2 = 6\Omega$ .

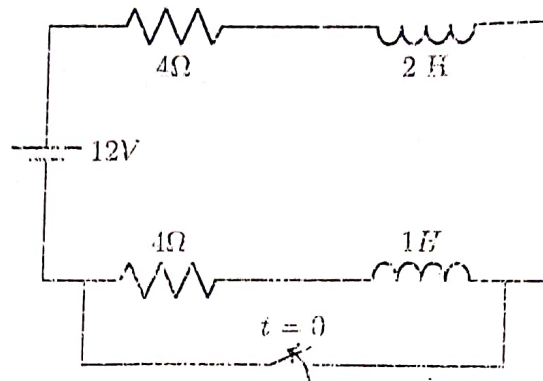


[3]

Exam Date: 2025-09-11

Total Time: 1.5 Hours Total Marks: 20

5. Determine the change of current through the 2H inductance with time, in the circuit shown below. Assume that the switch is opened at  $t = 0$ . [3]



6. Consider the Fermi Dirac distribution, where, the probability of occupying an energy state  $E$  is given by:

$$f(E) = \frac{1}{1 + e^{(E-E_f)/(k_B T)}}$$

Where:

- $E$ : The energy of the quantum state.
  - $E_f$ : The Fermi energy (or chemical potential), which is a fundamental thermodynamic quantity. At 0K, is the highest energy level occupied by an electron. At  $T > 0K$ , it is the energy level where the probability of occupation is exactly 0.5.
  - $k_B$ : Boltzmann constant ( $1.38 \times 10^{-23} \text{J/K}$  or  $8.617 \times 10^{-5} \text{eV/K}$ ).
  - $T$ : Absolute temperature in Kelvin.
- (a) For an intrinsic semiconductor show the electron and hole distribution in the respective bands at  $T=0$  and at  $T=300\text{K}$  (say). Show the position of the Fermi level explicitly.
- (b) How does the Fermi level shift in a n-type and a p-type semiconductor?
- (c) Draw the Density of states (DOS) as a function of energy in an intrinsic semiconductor. Do the DOS change as one increase the temperature from  $T=0$  to  $T=300\text{K}$ ?
- (d) In another diagram, show the electron and hole carrier density as a function of energy in an intrinsic semiconductor at room temperature.

[2+1+1+1]